MOSTAPHA EL ANSARI

Junior Machine Learning Engineer / **Sophia-Antipolis, France**

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EXPERIENCE

Bull SAS
Mar 2026 — Present

AI Engineer — Enterprise AI Sophia-Antipolis, FR

- Build multi-agent systems with AWS Strands, Bedrock AgentCore and LangGraph — inter-agent coordination, memory and tool orchestration.
- Deliver PoCs, PoVs and pre-sales demos for French enterprise clients, taking selected use cases to production on containerised AWS stacks.

Forvis Mazars Consulting
Oct 2025 — Feb 2026

AI Engineer (Internship) Casablanca, MA

- Built an LLM applicant-tracking prototype: Transformers CV parsing (NER, profile-to-job matching) and XGBoost scoring, served via FastAPI on Docker.
- Cut manual screening time by 60% and tracked experiments and model drift with MLflow.

Smartly R&D
Feb 2025 — Aug 2025

Machine Learning Engineer (Internship) Nice, FR

- Designed a real-time IoT ingestion pipeline (MQTT, PostgreSQL) and trained anomaly detection models (Isolation Forest, LSTM) in Docker.
- Deployed models behind FastAPI with Grafana monitoring for drift detection and automated alerting.

Locacoeur
Jun 2024 — Feb 2025

Data Engineer (Internship) Aix-en-Provence, FR

- Built an OCR/NLP pipeline (spaCy, PyTorch) for document extraction and a product classification API (Scikit-learn, FastAPI).
- Automated data migration and CRM sync, reducing manual processing from days to under 10 minutes.

PROJECTS

Real-Time Recommendation System — Two-Tower Retrieval
2025

PyTorch · Kafka · Redis · FAISS · Ray · Kubernetes

- Two-stage recommender over 2.1M items and 480k users: recall@50 0.23 → 0.38 via in-batch negatives with logQ popularity correction.
- Retrieval at 18ms p95 on FAISS HNSW indexes over 128-dim embeddings, refreshed hourly from a Kafka stream; validated by replayed counterfactual evaluation over 3 weeks of logged traffic.

Domain-Specific RAG System with Evaluation Harness
2025

Python · LangChain · Qdrant · sentence-transformers · Ragas

- QA over 12k regulatory documents at 2.4s p95; recall@5 61% → 84% with structure-aware chunking, hybrid search and reranking.
- Unfaithful answers 22% → 8% and query costs –38% via query decomposition, confidence-based abstention and model routing.

LLM Inference Optimization and Cost Analysis
2025

vLLM · PyTorch · CUDA · Docker · Prometheus/Grafana

- Self-hosted Llama-3.1-8B on one A10G: 9× throughput (95 → 870 tok/s) with continuous batching and PagedAttention.
- Evaluated AWQ 4-bit quantization and built a cost model — self-hosting wins above ~65% sustained GPU utilisation.

SUMMARY

Junior AI Engineer with experience building LLM applications, multi-agent systems, RAG, NLP and deep learning solutions for enterprise use cases.

Seeking a CDI or CDD opportunity to apply skills in Python, PyTorch, LangGraph, AWS, Docker, FastAPI and MLOps to production AI systems.

TECHNICAL SKILLS

PROGRAMMING
Python, R, SQL, Bash, VBA

ML & AI
PyTorch, Scikit-learn, Hugging Face Transformers, XGBoost, spaCy, NLP, Computer Vision, Deep Learning, Pandas, NumPy

LLMS & AGENTS
AWS Strands, Amazon Bedrock, Bedrock AgentCore, LangGraph, Multi-Agent Systems, RAG, ChromaDB, Qdrant

MLOPS, CLOUD & INFRA
AWS (S3, EC2, EKS, Bedrock, Lambda), Docker, Kubernetes, Pulumi, IaC, MLflow, GitHub Actions, CI/CD

OBSERVABILITY
LangSmith, Grafana, Prometheus, CloudWatch, ML/LLM monitoring, logging, metrics, tracing

SOFTWARE & DATABASES
FastAPI, PostgreSQL, Git, MQTT, Streamlit

EDUCATION

Ecole Centrale Casablanca
Sep 2022 — Aug 2026 · Casablanca, MA

Ingénieur Généraliste (MSc level) — AI, NLP, Deep Learning, Data Engineering

Ecole Centrale Marseille
Sep 2023 — Aug 2024 · Marseille, FR

Data Science exchange — Deep Learning, CNNs, Computer Vision

LANGUAGES

Arabic	Native
French	C1
English	C1